www.tntextbooks.in

Thus, when a number of resistors are connected in parallel, the sum of the reciprocals of the individual resistances is equal to the reciprocal of the effective or equivalent resistance. When 'n' resistors of equal resistances R are connected in parallel, the equivalent resistance is $\frac{R}{n}$.

i.e., $\frac{1}{R_P} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} \cdots + \frac{1}{R} = \frac{n}{R}$.

Hence, $R_p = \frac{R}{n}$

The equivalent resistance in a parallel combination is less than the lowest of the individual resistances.

4.7.3 Series Connection of Parallel Resistors

If you consider the connection of a set of parallel resistors that are connected in series, you get a series – parallel circuit. Let R_1 and R_2 be connected in parallel to give an effective resistance of R_{P1}. Similarly, let R_3 and R_4 be connected in parallel to give an effective resistance of R_{P2}. Then, both of these parallel segments are connected in series (Figure 4.8).

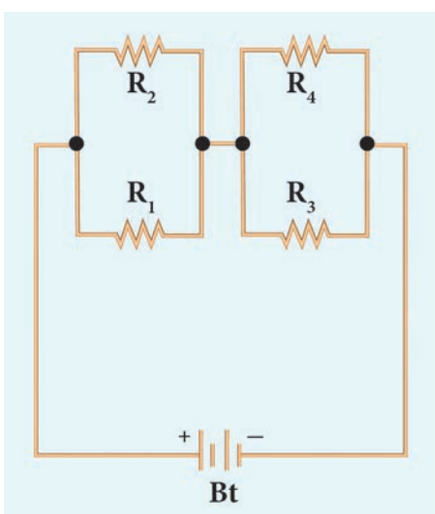


Figure 4.8 Series-parallel combination of resistors

Using equation (4.18), you get

$$\frac{1}{R_{P1}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{P2}} = \frac{1}{R_3} + \frac{1}{R_4}$$

Finally, using equation (4.12), the net effective resistance is given by $R_{total} = R_{P1} + R_{P2}$

4.7.4 Parallel Connection of Series Resistors

If you consider a connection of a set of series resistors connected in a parallel circuit, you get a parallel-series circuit. Let R_1 and R_2 be connected in series to give an effective resistance of R_{S1}. Similarly, let R_3 and R_4 be connected in series to give an effective resistance of R_{S2}. Then, both of these serial segments are connected in parallel (Figure 4.9).

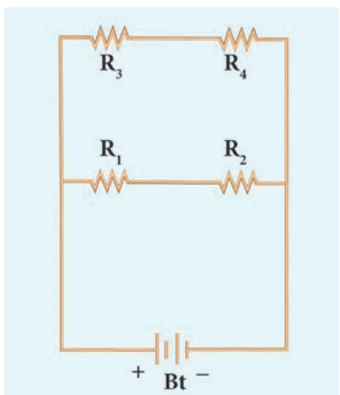


Figure 4.9 Parallel-series combination of resistors

Using equation (4.12), you get

$R_{S1} = R_1 + R_2$, $R_{S2} = R_3 + R_4$

Finally, using equation (4.18), the net effective resistance is given by

$$\frac{1}{R_{total}} = \frac{1}{R_{S1}} + \frac{1}{R_{S2}}$$

4.7.5 Difference between series and parallel connections

The difference between series and parallel circuits may be summed as follows in Table 4.3

10th_Science_Unit-4.indd 49 06-12-2021 18:29:12